Reinforcement Learning Assignment 2

Exercise 1

$$w_{t+1}(i) = \frac{w_t(i) \exp(\eta(-\ln p_{i,t}(x)))^{*}}{\sum_j w_t(j) \exp(\eta(-\ln p_{j,t}(x)))^{**}}$$

$$* \quad w_t(i) \exp(-\eta \ln(p_{i,t}(x))) = w_t(i) \exp[\ln(p_{i,t}(x)^{-\eta})]$$
$$= w_t(i)\, p_{i,t}(x)^{-\eta}$$

$$** \quad \sum_j w_t(j) \exp[-\eta \ln(p_{j,t}(x))]$$
$$= \sum_j w_t(j) \exp[\ln(p_{j,t}(x)^{-\eta})]$$
$$= \sum_j w_t(j)\, p_{j,t}(x)^{-\eta}$$

Should be i I guess or I don't understand

$$w_{t+1}(i) = \frac{p_{i,t}(x)^{-\eta} \cdot w_t(j)}{\sum_j p_{j,t}(x)^{-\eta} \cdot w_t(j)}$$

⚠ Sufficient? I Don't understand. Because the more the experts are wrong the more you trust them?